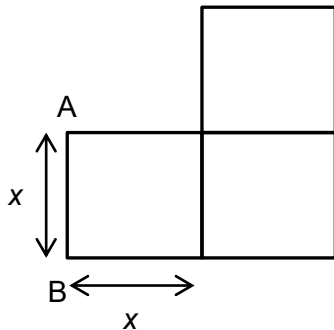


- 4** A uniform square metal sheet of length x is cut into an 'L' shape.

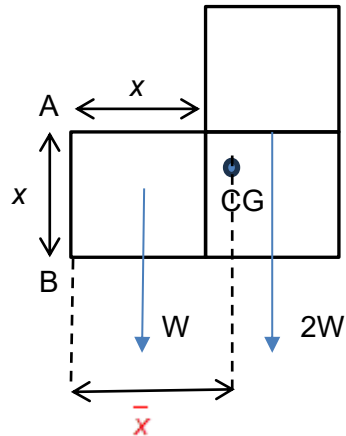


What is the distance of the centre of gravity of the sheet of metal from side AB?

- | | | | | | | | |
|----------|---------|----------|---------|----------|---------|----------|---------|
| A | $1.0 x$ | B | $1.2 x$ | C | $1.5 x$ | D | $1.8 x$ |
|----------|---------|----------|---------|----------|---------|----------|---------|

[Turn over

Answer: B



Let the weight of 1 square be W . Hence, the weight of 2 stacked squares on the right will be $2W$.

Let the horizontal distance of the CG of the 'L' shape be $\bar{x}$ from B.

Hence, the equivalent weight of the combined L shape will be $3W$ acting downwards on this CG.

Taking moments about B,

Moment of the combined weight of the L shape about B = Sum of moments of the weight of the individual squares about B.

$$3W(\bar{x}) = W(0.5x) + 2W(1.5x)$$

$$3\bar{x} = 3.5x$$

$$\bar{x} = 1.2x$$

5	Several identical springs each having the same spring constant are joined in four arrangements
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